

Scan2Plan CPQ

Handoff Package for Elijah Tuttle

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March 26, 2026

Owen declined the partnership on March 25, 2026. This package covers infrastructure, data, and product documentation so Elijah can operate the Scan2Plan CPQ independently.

Live Product: <https://pipeline.scan2plan.dev/>

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Part I: What You're Getting (and What You're Not)

Transfers to Elijah

- **GCP project access** (s2px-production) -- IAM roles for managing infrastructure
- **GCS scan data and project files** -- all business data in the storage buckets
- **Database access** -- Cloud SQL connection for the production PostgreSQL instance
- **Domain** (scan2plan.io) -- can be transferred if desired
- **Product documentation** -- this document and related specs
- **Firebase Hosting sites** -- the deployed frontend URLs

Does NOT Transfer

- **Source code** -- The CPTV27/S2PX GitHub repository is Chase's private IP. No codebase leaves.
- **GitHub access** -- All collaborator access was revoked on March 25, 2026.
- **S2PX platform** -- S2PX is Chase's separate AI-first operating system. It is a completely different product from the Scan2Plan CPQ.
- **Any HDX / Measurably Better / Hillbilly Dreams IP** -- completely separate.
- If the CPQ source code is needed, that is a separate licensing conversation.

Part II: Product Overview

The Scan2Plan CPQ is a full-lifecycle platform for 3D scanning and BIM businesses. It connects lead intake through scoping, pricing, proposals, signatures, and production delivery in one system.

Feature	Description
Sales Pipeline	Kanban board / lead management with stage tracking
Scoping Forms	135-field intake form with geocoding, multi-building support
Deal Workspace	Full lead detail view with quote builder, documents, engagement tracking
CPQ Calculator	Multi-building, multi-discipline pricing engine with margin controls
Production Pipeline	Project tracking from scan through delivery with blocker enforcement
Proposal Builder	WYSIWYG proposal editing with versioning and PDF export
Signature Flow	Dual e-signature with audit trail and signed PDF generation
Scan Tech Portal	Field technician mobile interface with photo analysis AI
QuickBooks Integration	Estimate/invoice sync via OAuth2
Google Drive Integration	File management via GCS 4-bucket architecture
Knowledge Base	Internal wiki with AI Q&A chat
Strategy Center	CEO strategy drafts with 3-tier governance
AI Agents (7)	Pricing, Operator, Photo Analysis, Checklist, Audio-to-Scoping, KB Chat, Strategy Advisor
Scorecard	Company performance metrics and financial dashboards
Website Manager	Content management for scan2plan.io marketing site

Note on the CPQ Calculator

The pricing engine was the single most labor-intensive component of the system. The underlying pricing logic changed repeatedly during development, requiring multiple full refactors. The current engine is the stabilized version, backed by database-driven rate tables (15 pricing tables) and covered by 261 unit tests.

Part III: GCP Infrastructure

Field	Value
Project ID	s2px-production
Owner Account	chase@scan2plan.io
Service Account	s2px-api@s2px-production.iam.gserviceaccount.com
Region (primary)	us-east4 (N. Virginia)
Region (sandbox)	us-central1 (Iowa)

Services in Use

- Cloud SQL (PostgreSQL)
- Cloud Run (API backend)
- Cloud Storage (4 buckets)
- Firebase Hosting (3 targets)
- Firebase Auth (Google OAuth)
- Secret Manager
- Cloud Tasks
- Cloud CDN (signed URLs)

Part IV: Google Cloud Storage

Bucket	Purpose	Env Var
s2p-core-vault	Primary storage -- proposals, certificates, exports	GCS_BUCKET
s2p-active-projects	Active project data -- point clouds, tours, deliverables	GCS_ACTIVE_BUCKET
s2p-incoming-staging	Upload staging area -- files land here before processing	GCS_STAGING_BUCKET
s2p-quarantine	Quarantine bucket for failed/suspicious uploads	Security allowlist

What's in the Buckets

- Scan data (.e57, .las, .laz point cloud files)
- Virtual tour assets (equirectangular panoramas, floor plan SVGs)
- Proposals (generated PDFs)
- Certificates of Assurance (generated PDFs)
- Field photos (scan tech uploads)
- Project deliverables (BIM files, CAD exports)

All of this is Scan2Plan business data and can be transferred.

GCS Folder Structure (inside s2p-active-projects)

```
{ProjectFolder}/
project.json
Deliverables/
Point Cloud/
{raw-files} (.las, .laz, .e57, .rcp, .rcs)
potree/ (metadata.json, octree.bin, hierarchy.bin)
Virtual Tour/
tour-manifest.json, floorplan.svg, panoramas/
BIM/CAD/
Photos/
Documents/
```

Auto-Processing Pipelines

Files uploaded to GCS are processed automatically by Cloud Run workers:

Worker	Input	Output	Trigger
Virtual Tour	ZIP bundle in staging	tour-manifest.json + panoramas	POST /api/projects/:id/virtual-tour/process
Point Cloud	Raw .e57/.las/.laz	Potree octree files	Manual worker invocation
Sidecar Writer	Project data changes	project.json metadata	Automatic on stage transitions

Part V: Firebase & Hosting

Target	Firebase Site	URL	API Service
production	s2px	s2px.web.app	s2px-api (us-east4)
sandbox	s2px-sandbox	s2px-sandbox.web.app	s2px-api-sandbox (us-central1)
staging	s2px-staging	s2px-staging.web.app	s2px-api-staging (us-east4)

Firebase Services

- **Auth** -- Google OAuth provider (only auth method)
- **Hosting** -- Static SPA hosting with API rewrites to Cloud Run
- **Storage Rules** -- defined in storage.rules
- **Firestore** -- Client-side only, supplementary to Cloud SQL (see below)

Firestore Collections (Not Finalized)

Firestore was being evaluated as a supplementary data layer alongside Cloud SQL. Only wiki_pages is confirmed active. The deal_* collections were not finalized. The primary database for all business data is Cloud SQL (PostgreSQL).

Collection	Purpose	Status
wiki_pages	Knowledge Base articles (legacy)	Active
deal_sessions	Viewer session telemetry	Not finalized
deal_events	Section enter/exit tracking	Not finalized
deal_voice	Voice memo transcriptions	Not finalized
deal_cta_clicks	CTA click tracking	Not finalized

WARNING: All three environments (production, staging, local) share the same Firebase Auth project. There is no auth isolation between environments.

Part VI: Database

Field	Value
Type	PostgreSQL (Google Cloud SQL)
Instance	s2px-production-db
ORM	Drizzle ORM
Connection	Via DATABASE_URL env var
Max Connections	~100 (Cloud SQL instance setting)
Pool Size	5 per Cloud Run instance
Backups	Daily automated, 7-day retention + PITR enabled

Schema (~90 tables across 8 files)

File	Tables	Contents
shared/schema/db.ts	59 core tables	users, scoping_forms, quotes, proposals, production
shared/schema/pricing.ts	15 tables	Rate tables, baselines, modifiers
shared/schema/platform.ts	2 tables	Content, roadmap
shared/schema/market-intelligence.ts	9 tables	Segments, competitors, scanners
+ 4 additional files	5+ tables	Deal arena, engagement, AI usage, initiatives

Key Tables

Table	Purpose
users	Firebase-authed users with RBAC roles
scoping_forms	Deal intake (135 columns)
quotes	Generated quotes with line items
proposals	Proposal documents with PDF metadata
production_projects	Active production pipeline
qbo_tokens	QuickBooks OAuth tokens
pricing_constants	V2 pricing engine config

Part VII: Cloud Run Services

Service	Region	Purpose
s2px-api	us-east4	Production API
s2px-api-sandbox	us-central1	Sandbox API
s2px-api-staging	us-east4	Staging API

Production Configuration

Setting	Value
Max instances	10 (capped for connection safety)
Concurrency	80
Pool max	5 per instance

Connection pool rule: instances x pool_max must be <= 0.7 x cloud_sql_max_connections

Part VIII: Environment Variables & Secrets

Required

Variable	Purpose
DATABASE_URL	PostgreSQL connection string
GCS_BUCKET	Primary GCS bucket (s2p-core-vault)
VITE_FIREBASE_*	6 Firebase Web SDK config values
FIREBASE_PROJECT_ID	Server-side Firebase project ID
GEMINI_API_KEY	Google Gemini AI (powers all 7 agents)
GOOGLE_MAPS_API_KEY	Server-side Maps API
VITE_GOOGLE_MAPS_API_KEY	Client-side Maps API

Optional

Variable	Purpose
ANTHROPIC_API_KEY	Claude AI (KB extraction)
QBO_CLIENT_ID / SECRET	QuickBooks Online OAuth2
SENDGRID_API_KEY	Email delivery (proposals)
ENCRYPTION_KEY	Encrypts stored API keys in DB

Variable	Purpose
GCS_ACTIVE_BUCKET	Active projects bucket
GCS_STAGING_BUCKET	Upload staging bucket

Secrets are stored in GCP Secret Manager on the s2px-production project. Cloud Run services pull secrets at deploy time.

Part IX: Domains & DNS

Domain	Purpose	Notes
scan2plan.io	Primary domain	Can be transferred
pipeline.scan2plan.dev	CPQ product URL	Live product
s2px.web.app	Production Firebase Hosting	Tied to Firebase project
s2px-sandbox.web.app	Sandbox	Tied to Firebase project
s2px-staging.web.app	Staging	Tied to Firebase project

Part X: Third-Party Integrations

Service	Purpose	Account
Firebase	Auth + Hosting	chase@scan2plan.io
Google Cloud Storage	4-bucket file architecture	s2px-production
Google Maps	Geocoding, travel calc	API key on project
Google Gemini AI	7 AI agents (server-side)	API key
QuickBooks Online	Estimate/invoice sync	Owen's QBO -- needs disconnect
SendGrid	Proposal email delivery	API key
Cloud Tasks	Async job processing	s2px-production
Cloud CDN	Secure file delivery (signed URLs)	s2px-production

QuickBooks Note: The QBO integration connects to Owen's QuickBooks account. Since Owen is out, this connection needs to be either disconnected (revoke OAuth tokens) or reconnected to a different QBO account.

Part XI: Auth Architecture

- **Provider:** Firebase Auth with Google OAuth (only auth method)
- **Flow:** Client-side Google sign-in -> Firebase token -> Bearer token in every API call -> server verifies and attaches req.user
- **Roles:** developer, admin, ceo, production, scantech, viewer
- **Frontend guard:** RequireAuth component with allowedRoles
- **Backend guard:** requireRole() middleware on each route
- **No password auth** -- Google OAuth only

Part XII: How to Get This Running

Step 1: Get GCP Access

Chase adds your Google account to s2px-production with IAM roles:

- roles/cloudsql.client (database access)
- roles/storage.admin (GCS bucket access)
- roles/run.admin (Cloud Run management)
- roles/firebase.admin (Firebase hosting + auth)
- roles/secretmanager.secretAccessor (read secrets)

Step 2: Get Credentials from Chase

Chase provides DATABASE_URL, Firebase config values, GEMINI_API_KEY, Google Maps keys, and any optional keys needed.

Step 3: Verify Access

```
gcloud auth login
gcloud config set project s2px-production
gcloud sql instances list
gcloud storage buckets list
gcloud run services list
```

Note: Running the application locally or deploying updates requires the source code, which is not part of this handoff. The steps above verify infrastructure access only.

Part XIII: Credentials Checklist

Chase fills in and delivers separately (never in this document):

- ☐ GCP IAM access granted to Elijah's Google account
- ☐ DATABASE_URL connection string
- ☐ Firebase config block (6 VITE_FIREBASE_* values)
- ☐ FIREBASE_PROJECT_ID
- ☐ GEMINI_API_KEY
- ☐ GOOGLE_MAPS_API_KEY (server)
- ☐ VITE_GOOGLE_MAPS_API_KEY (client)
- ☐ ENCRYPTION_KEY (if applicable)
- ☐ SENDGRID_API_KEY + SENDGRID_FROM_EMAIL (if email delivery needed)
- ☐ ANTHROPIC_API_KEY (if Claude AI features needed)
- ☐ Domain registrar transfer (if scan2plan.io is being transferred)
- ☐ QBO decision: disconnect Owen's account or reconnect to new account

Part XIV: Ownership Boundary

Asset	Owner	Transfers?
Source code (CPTV27/S2PX repo)	Chase Pierson	NO
GitHub repository access	Chase Pierson	NO
S2PX platform (separate product)	Chase Pierson	NO
HDX / Measurably Better / Hillbilly Dreams IP	Chase / HDX	NO
GCS scan data & project files	Business data	YES
GCP project access (s2px-production)	Chase (owner)	YES -- IAM roles
Cloud SQL database & data	Business data	YES
Firebase Hosting sites	Tied to GCP project	YES
Firebase Auth user accounts	Tied to Firebase project	YES
Domain (scan2plan.io)	Chase	YES -- can transfer
Product documentation	General knowledge	YES
API keys	Account-specific	CONDITIONAL -- re-provision

Clear Boundaries

- **The code is Chase's.** The CPTV27/S2PX repository is private and stays with Chase. No source code is part of this handoff.
- **S2PX is a separate product.** S2PX is Chase's AI-first operating system platform. The Scan2Plan CPQ is a focused product that was built for the scanning business. They are different systems.
- **The data is the business's.** GCS files, database records, and project deliverables are Scan2Plan operational data.
- **Infrastructure access is grantable.** Chase can add Elijah to the GCP project with IAM roles without transferring ownership.
- **API keys should be re-provisioned.** Elijah should create his own Google Cloud project and API keys long-term.

Part XV: Product Documentation Summary

A separate 13-page product deck (Scan2Plan_App_Deck.pdf) documents each feature with screenshots. Summary:

Feature	Description
Sales Pipeline	Kanban deal management with value, priority, source, and win probability
Deal Workspace	Central record for CRM data, billing, project info, and downstream actions
CPQ Calculator	Multi-area, multi-discipline quoting with version history, travel, and risk logic
Production Pipeline	Operations board: Scheduling through Delivered with blocker enforcement
Proposal Builder	Versioned WYSIWYG proposal editing and export-ready PDF generation
Signature Flow	Sender-first or direct client signature with public signing page and signed PDF output
ScanTech PDF Generator	Mission brief PDF for field execution with location, scope, conditions, equipment
QuickBooks + Google Drive	Estimate sync, folder creation, document uploads, project file organization
Tech Stack	React, TypeScript, Vite, Tailwind, Express, PostgreSQL, Drizzle ORM, Vitest, Playwright
Integration Map	Lead -> CPQ Quote -> Proposal -> Signature -> Closed Won -> Production Project

Roadmap / What's Next

- Finish proposal builder stabilization and PDF parity hardening
- Tighten signature completion notifications
- Continue Drive/shared-folder reliability work
- Expand production UX exposure in the trimmed build
- Keep reducing handoff friction between sales and operations

Part XVI: Development Timeline

Based on the full git history. Development ran from January 19 to February 15, 2026 -- 28 days of active development.

Phase	Dates	Focus	Key Deliverables
Phase 1	Jan 19-20	Foundation	Initial repo baseline, Sales & Production direction established
Phase 2	Jan 21-25	Sales Pipeline & Proposals	Pipeline UX, proposal versioning, WYSIWYG editor, signature links
Phase 3	Jan 29	CPQ Hardening	Pricing fixes, custom line items, pricing-summary grouping
Phase 4	Feb 2	Dual Signature & Polish	Dual-signature flow, audit trail, PDF typography, discipline rollup
Phase 5	Feb 4-10	Pricing Controls	Tier A overrides, Quote Builder behavior, theme toggle, proposal editab
Phase 6	Feb 15	Drive Automation	Google Drive uploads, shared Drive support, proposal editor rebuild

Current Product State

Shipped:

- Sales pipeline
- Deal workspace
- Versioned CPQ quoting
- Proposal versioning
- WYSIWYG proposal editor
- PDF export
- Sender and client signature workflow
- Production tracker
- Mission brief PDF generation
- QuickBooks estimate integration
- Google Drive folder and upload flows

Still in progress:

- Proposal builder stabilization and parity work
- Post-signature notification completion
- Drive reliability hardening
- Some trimmed-build navigation and production UX exposure

The last commit in the CPQ repository is dated February 15, 2026. Development paused after the Drive automation and editor rebuild delivery cluster.

Part XVII: Product Manual Summary

A separate 7-page product manual (Scan2Plan_App_Manual.pdf) provides detailed how-to instructions for each feature. The end-to-end usage pattern:

1. Create the deal in the Sales Pipeline
2. Complete lead and billing data in the Deal Workspace
3. Build the quote in CPQ
4. Create the proposal version and edit it
5. Export the PDF and send the proposal
6. Complete sender signature if required
7. Collect the client signature
8. Confirm the deal closes won and hands off to production
9. Use the Mission Brief and production pipeline to execute the work

Known Limitations

- **Signature flow:** Post-signature outbound email notification not fully wired. Token expiry requires link regeneration.
- **Proposal builder:** Under active stabilization. Multiple Feb commits focused on save reliability and PDF parity.
- **Navigation:** Some routes exist that aren't exposed in the trimmed sidebar.
- **Google Drive:** Folder creation is non-blocking; projects can still be created if Drive fails.
- **QuickBooks:** Requires valid OAuth credentials and connected company before estimate operations work.

Part XVIII: Financial Notes

Chase's 5% development arrangement ended in February 2026. No charges from Chase for March 2026.

Owen is responsible for paying Elijah for any development time in February and March 2026.

Accompanying Documents

- **Scan2Plan_App_Deck.pdf** -- 13-page product overview with screenshots
- **Scan2Plan_Project_Timeline.pdf** -- Git-history-based development timeline
- **Scan2Plan_App_Manual.pdf** -- Full product manual with how-to instructions

This document was generated from codebase configuration, GCP project settings, and architecture documentation. GCS bucket inventory should be verified live with gcloud storage buckets list once access is granted.